

9 (short)

(a) The molecular structure of crystalline polyethylene is shown in Fig. 7. Calculate the density of this polyethylene structure, and explain whether this value is an upper or lower bound for practical polyethylenes. [4]

(b) Would the following polymers be used mostly above or below their glass transition temperature? Justify your answer with reference to the molecular structure, and its influence on their properties.

- (i) Semi-crystalline thermoplastic.
- (ii) Amorphous thermoplastic.
- (iii) Natural rubber.

[6]

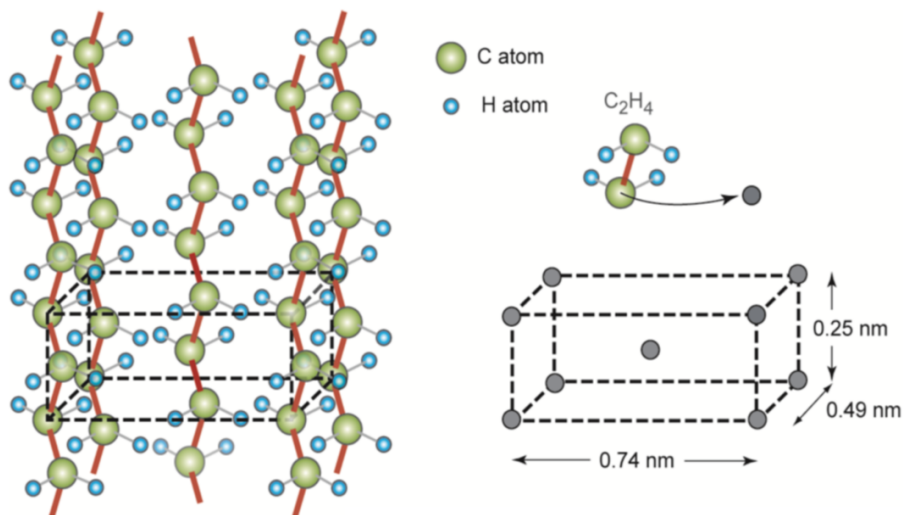


Fig. 7